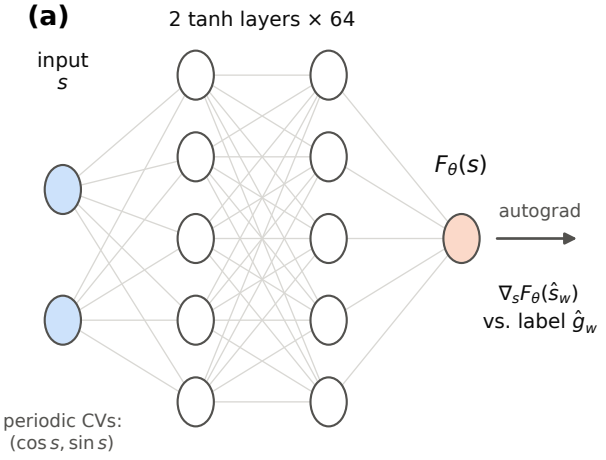
(a)
2 tanh layers × 64
input
s
$F_\theta(s)$
autograd
$\nabla_s F_\theta(\hat{s}_w)$
vs. label $\hat{g}_w$
periodic CVs:
(cos s, sin s)
Loss $\mathcal{L}_m = \sum_w \omega_{mw} |\nabla_s F_{\theta_m}(\hat{s}_w) - \tilde{g}_{mw}|^2 + \lambda \|\theta_m\|^2$
(b)
Random init
same labels; members differ only in initial weights $\theta_m^{(0)}$
Bootstrap
ω_{mw} = resampling counts of windows (with replacement)
Noise-injected
labels $\hat{g}_w + \hat{\varepsilon}_w \xi_{mw}$, $\xi \sim \mathcal{N}(0,1)$; weights $\omega_{mw} \propto \hat{\varepsilon}_w^{-2}$
Gradient GP
exact posterior; label noise $\hat{\varepsilon}_w^2$ on the diagonal